Isolating Smart TVs and IoT Devices with UniFi

A tested, dual-stack VLAN, DNS filtering and firewall-zone walkthrough

Test platform: UDM-SE | UniFi OS 6.0.15 | UniFi Network 11.0.94 | LG Smart TV | Linux test laptop

1. Objective and topology

Allow a smart TV to reach the Internet without unrestricted access to trusted computers and NAS devices. The same architecture can be adapted to other IoT equipment, with exceptions where required.

Network	VLAN	IPv4 subnet	Purpose
PepperNet / Default	1	10.0.1.0/24	Trusted devices
Hobble Skirt	107	10.0.107.0/24	LG TV / isolated devices

Key lesson: A VLAN alone is not isolation. The zone-based firewall policy determines whether devices can communicate across networks.

2. Create the Hobble Skirt network

In UniFi Network, go to **Settings > Networks** and create the network with VLAN ID **107**, IPv4 gateway **10.0.107.1/24**, DHCP enabled and Internet access enabled. Enable IPv6 with a separate ISP-delegated /64, SLAAC and Router Advertisements. Use your own IPv6 prefix, not the example addresses in this guide.

The **Isolate Network** checkbox was left off in the tested configuration because the dedicated firewall zone provided isolation.

3. Create and assign a Wi-Fi SSID

Under **Settings > WiFi**, create an IoT SSID and assign it to Hobble Skirt (VLAN 107). Confirm the relevant access point broadcasts it. Connect the television. The TV was given a reserved IPv4 address of **10.0.107.107**; a reservation is convenient, not mandatory.

4. Create the Armbinder firewall zone

In the Zone-Based Firewall settings, create a zone called **Armbinder** and assign only Hobble Skirt to it. Leave PepperNet / Default in **Internal**. Inspect the zone matrix: in the tested setup the new zone could access External and Gateway but was blocked from other internal zones.

Traffic	Intended policy
Armbinder to External	Allow Internet access
Armbinder to Internal	Block by default
Internal to Armbinder	Block unless explicitly needed
Armbinder to Gateway	Allow only necessary gateway services

Defaults may vary. Inspect the actual policy matrix and rule ordering. Gateway access must support necessary DNS, DHCP and IPv6 control traffic.

5. Create the Ball Gag DNS filter

Create a UniFi **Content Filtering** policy named **Ball Gag**, with the **entire Hobble Skirt network** as its source. The following custom blocklist was used for the LG television:

```
eu.info.lgsmartad.com
config-overlay-ads.alphonso.tv
eu-mercury-lb.alphonso.tv
securepubads.g.doubleclick.net
sb.scorecardresearch.com
eic.ads.lgtvcommon.com
gb.ad.lgsmartad.com
```

These are LG-specific examples, not a universal IoT blocklist. Avoid broad manufacturer or CDN blocks without testing. DNS filtering is not a complete egress-control solution: alternative DNS, encrypted DNS and direct-IP connections may need separate controls.

6. Test DNS filtering over IPv4 AND IPv6

Connect a Linux laptop to the Hobble Skirt SSID. Query the gateway DNS listeners directly:

```
dig @10.0.107.1 eu.info.lgsmartad.com A
dig @2a02:8013:785f:1::1 eu.info.lgsmartad.com AAAA
```

Replace the IPv6 gateway with your own. On the tested system, filtered answers returned A **203.0.113.250** and AAAA **2001:db8:1000::fa**. Other versions may use different responses; the key is consistent blocking on both protocols.

Version-specific finding: IPv6 gateway DNS filtering was bypassed on the tested UniFi OS 6.0.13 / Network 11.0.94 configuration. Updating **UniFi OS to 6.0.15** fixed the observed bypass without changing the Network application version. An earlier Network 10.x configuration had also exhibited this issue. This does **not** mean all IPv6 firewalling was broken.

7. Test routes and actual connectivity

```
ip -4 route get 1.1.1.1
ip -6 route get 2606:4700:4700::1111
ping -c 4 1.1.1.1
ping -6 -c 4 2606:4700:4700::1111
```

Check that the chosen source addresses belong to the IoT VLAN. Route lookups show routing decisions; successful pings demonstrate packet delivery. Both IPv4 and IPv6 Internet connectivity succeeded during testing.

8. Test isolation from the trusted network

From the IoT SSID, attempt to reach known trusted-LAN devices, for example:

```
ping -c 4 10.0.1.150
```

Repeat against a known trusted device global IPv6 address. Before creating Armbinder, trusted devices were reachable from the TV VLAN; afterwards the same IPv4 and IPv6 pings failed. **Ping failure alone is not a security audit:** also check relevant TCP services and firewall policy/logs.

9. Configure gateway mDNS proxy

In **Gateway mDNS Proxy**, select **Custom**. The tested selection was **Default (1)** only under VLAN Scope, leaving **Hobble Skirt (107)** unchecked; Service Scope was **All**. The intention is to avoid proxying cross-VLAN discovery into the TV network. This does not disable same-VLAN discovery and does not substitute for firewall rules.

If AirPlay, Chromecast, HomeKit or a local controller must operate across VLANs, discovery and narrowly scoped firewall exceptions may be needed.

10. Optional: Plex, Emby and NAS access

If a TV must reach a trusted media server, add a narrowly scoped allow rule from the TV or IoT VLAN to the server IP and required service ports, ahead of the relevant deny policy. Stateful return traffic for an allowed connection is normally permitted. Automatic discovery may require separate mDNS or multicast configuration. **This exception was not implemented or tested in the original setup.**

11. Other IoT devices: consider dependencies

Smart locks, thermostats, hubs and controllers may require local discovery, cloud access or communication with a trusted controller. Determine which device initiates each connection, which addresses/ports are necessary, and what happens if the Internet is unavailable. For security-critical devices, validate normal operation and failure behaviour before depending on the configuration.

12. Final checks

- Correct VLAN address assignment and DHCP
- IPv4 and IPv6 Internet connectivity
- Gateway DNS filtering on A and AAAA queries
- Isolation from trusted devices over both protocols
- Streaming and device-specific features still function
- Expected mDNS discovery behaviour
- Only narrowly scoped, documented exceptions

Verification note: BBC and Apple streaming were confirmed working *before* the final firewall-zone and mDNS changes. Re-test those services on the finished configuration before claiming end-to-end application verification.

The naming convention

Hobble Skirt (VLAN): restricts where the TV can wander. **Ball Gag** (DNS filter): stops it talking to unwanted services. **Armbinder** (firewall zone): stops it poking around the trusted network.

The practical lesson: configure the firewall deliberately, test both IP protocols, and verify traffic behaviour instead of trusting labels in the UI.